Adrian Franco

Adrian Franco Preamp calculations:

Gain stage

Gain:

$$G = 1 + \frac{R_F}{R_G}$$

$$1 + \frac{100k}{10k} = 11$$

Gain(dB):

$$G = 20 \log 11$$

$$20 \log 11 = 20.8\ dB$$

Frequency of High pass corner (gain stage):

$$f(Hz) = \frac{1}{2\pi RC}$$

$$\frac{1}{2\pi \times 10k \times 1nF} = 1.59\ Hz$$

Frequency of Low pass corner (gain stage):

$$f(Hz) = \frac{1}{2\pi RC}$$

$$\frac{1}{2\pi \times 10k \times 47\mu F} = 0.338\ Hz$$

Active Baxandall circuit

Bass side resistors (Rpot = 100k)

$$G = \frac{R_2}{R_1}$$

$$\frac{(100k + R)}{R} = 4$$

$$4R = 100k + R$$

$$R = \frac{100k}{3}$$

$$R \approx 33k$$

Capacitor values for low corner (bass), target 50 Hz:

$$f = \frac{1}{2\pi RC}$$

$$C = \frac{1}{2\pi fR}$$

$$C = \frac{1}{2\pi \times 33k \times 50}$$

$$C = 31.8nF \rightarrow 33nF$$

Bass Gain range:

$$G = \frac{100k + 33k}{33k}$$

$$G \approx 4$$

$$G = 20\log(4) = 12dB$$

$$G = \frac{33k}{(100k + 33k)}$$

$$G = 0.248$$

$$G = 20\log 0.248$$

$$G = -12dB$$

$$G = \pm 12dB$$

Treble side resistor:

$$1 + \frac{33k}{R} = 4$$

$$3 = \frac{33k}{R}$$

$$R = 11k$$

Capacitor (target 3k Hz)

$$C = \frac{1}{2\pi \times 11k \times 3k}$$

$$C = 4.82nF \rightarrow 4.7nF$$

Treble gain range:

$$G = 1 + \frac{33k}{11k} = 4$$

$$G = 20 \log 4 = 12dB$$

$$G = \frac{\frac{1}{33k}}{\frac{1}{33k} + \frac{1}{11k}}$$

$$G \approx 0.25$$

$$G = 2 \log 0.25 = -12dB$$

$$G = \pm 12dB$$